Vocal Volume Macro — User Manual

> *Melodyne-style vocal dynamic correction for REAPER*
> `VocalVolumeMacro_v6.lua` *— current production version*

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1. What this script does

The problem with compressors on vocals

A compressor reacts to amplitude — it has no awareness of what the sound *is*. When a loud **S** or **T** hits, the compressor pulls everything down, including the vowels around it. When a loud vowel hits, sibilants ride up with it.

Result: de-essers fight the compressor, the voice sounds over-processed, and natural articulation disappears.

What Melodyne does differently

Melodyne's Volume Macro analyses each "blob" (a detected note or noise event) and lets you scale the volume of tonal blobs only. Sibilants and plosives are left alone because they're already separated into their own blobs.

What this script does

This script reproduces that logic inside REAPER without Melodyne:

1. Splits the recording into 10 ms blocks
2. Runs four acoustic measurements on each block: RMS, ZCR, Spectral Centroid, Spectral Flux
3. Classifies each block as **tonal** (vowel) or **non-tonal** (consonant, sibilant, noise, silence)
4. Merges tonal blocks into **segments** — with configurable gap tolerance for vibrato handling
5. Applies volume correction **only to tonal segments**, respecting a gate threshold
6. Writes the result to the item's **Take Volume Envelope** — non-destructive, follows the item if you move it, armed automatically on write

After analysis, the **Target RMS** slider automatically aligns to the detected average level of the voice — no manual guessing needed.
What it does not do:

- Does not replace a compressor or limiter
- Does not correct pitch
- Does not work in real time (analyses then writes a static automation)
- Does not detect individual notes like Melodyne

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2. Installation

Requirements

Dependency	Version	Where to get it
REAPER	6.0 or later	reaper.fm
ReaImGui	latest stable	ReaPack → Browse packages → search "ReaImGui"

Installing ReaImGui

1. Open REAPER
2. **Extensions** → **ReaPack** → **Browse packages**
3. Search for **ReaImGui**
4. Right-click → **Install** → **Apply**
5. Restart REAPER

> *Without this step the script shows an error message and refuses to start.*

Installing the script

1. Download `VocalVolumeMacro_v6.lua`
2. Copy to your REAPER Scripts folder:
   - **Windows:** `C:\Users[you]\AppData\Roaming\REAPER\Scripts`

    - **macOS:** `~/Library/Application Support/REAPER/Scripts/`
3. In REAPER: **Actions → Show action list**
4. Click **New action → Load ReaScript**
5. Select the `.lua` file
6. Optional: assign a keyboard shortcut

First launch

Run via **Actions → Run script** or your shortcut. The **Vocal Volume Macro v6.0** window appears.

> ***Tip:*** *Open* ***Actions → Show REAPER console*** *before running. All diagnostic messages appear there — essential when troubleshooting.*

Settings persistence

All parameters are automatically saved between sessions. Your last settings are restored on every launch. No preset management needed.

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3. Interface overview

```
┌───────────────────────────────────────────┐
  │  Target item                              │  which take is being processed
  ├───────────────────────────────────────────┤
  │  Analysis parameters                      │  how tonal / noise is detected
  ├───────────────────────────────────────────┤
  │  Correction parameters                    │  how much gain is applied
  ├───────────────────────────────────────────┤
  │  [ Analyse ]  [ Apply ]  [ Reset env. ]   │  action buttons
  ├───────────────────────────────────────────┤
  │  Results & segment map                    │  stats after analysis
  └───────────────────────────────────────────┘
```

Target item section

Display	Meaning
✓ green	A valid audio item is detected and ready
✗ red	No item selected, or selected item has no audio source
[locked]	Item is frozen during/after analysis

> *The script validates the item before starting. It refuses with a clear error if the item is offline (file missing from disk), empty (MIDI or text take), or has no audio source.*

Once you click **Analyse**, the item is locked to prevent accidental deselection.

Progress bar

Appears during analysis. Shows block / total, percentage, and an **estimated time remaining** (displayed after 2% progress). The **Cancel** button interrupts cleanly.

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4. Parameters reference

Analysis parameters

These control *how* tonal blocks are detected. They do not change the gain — only the classification. You need to **re-analyse** after changing them.

> *All settings are saved automatically on every change.*

ZCR threshold

Default: 0.15 — Range: 0.05 to 0.40

Zero Crossing Rate: how often the signal crosses the zero axis per unit of time.

- Voiced vowels: **low, regular** ZCR
- Sibilants, fricatives and noise: **high, erratic** ZCR

Setting	Effect
0.05 – 0.10	Strict — only the cleanest vowels
0.12 – 0.18	**Recommended range for most voices**
0.20 – 0.40	Permissive — includes soft fricatives ("v", "z")

Spectral centroid max (Hz)

Default: 2500 Hz — Range: 200 Hz to freq_max

The "centre of gravity" of the frequency spectrum. Low-frequency rumble and plosive bass are filtered out before this measurement, so only the musically relevant spectrum matters.

- Sung vowels concentrate energy in the low-mid range
- Sibilants and noise push energy higher

> *After analysis, the slider shows `← max analysed: XXXX Hz` in yellow — the physical DFT limit. The slider is clamped to this value automatically.*

Consonant onset sensitivity (Flux)

Default: 0.30 — Range: 0.05 to 1.00

Measures abrupt spectral changes between consecutive blocks. Plosives (t, k, p, b, d) produce a sharp onset even when ZCR is not high.

Setting	Effect
0.05 – 0.15	Very sensitive — most plosives preserved
0.25 – 0.35	**Recommended starting point**
0.50 – 1.00	Less sensitive — more aggressive correction near consonants

Analysis resolution (ms/block)

Default: 10 ms — Range: 5 to 30 ms

Duration of each analysis block. Changing this value **invalidates the current analysis** — the GUI resets automatically.

Setting	Use case
5 ms	Maximum precision, slower, more envelope points
10 ms	**Standard — good balance**
20 – 30 ms	Faster, coarser — useful on very long takes

Correction parameters

These control *how much* gain is applied. You can change them **without re-analysing** — the preview updates immediately.

Target RMS level (dB)

Default: auto-set after analysis — Range: -40 to -6 dB

The level towards which tonal blocks or segments are corrected.

After each analysis, this slider **automatically aligns to the average RMS of the tonal blocks** — the true natural level of the singing voice in this recording, ignoring silences and noise. The stats panel shows `Auto-target ← auto` to confirm.

If you adjust the slider manually, the ↺ **Auto** button appears to restore the calculated value instantly.

> ***To verify the script is working:*** *if the stats show `Avg RMS: -18.5 dB` and your target is `-18 dB`, the correction is only +0.5 dB — barely visible on the envelope. Set the target to `-12 dB` or `-24 dB` to see clear envelope movement and confirm everything is working.*

Voice type	Typical auto-target range
Lead vocal (pop / rock)	-20 to -14 dB
Lead vocal (classical / lyric)	-24 to -16 dB
Rap / spoken word	-18 to -12 dB
Background vocals / choir	-28 to -18 dB
Acoustic / folk	-26 to -18 dB

Gate threshold (dB)

Default: -40 dB — Range: -60 to -10 dB

Blocks or segments **below** this level are left at unity gain (1.0) — no correction applied. This prevents the script from boosting breaths, microphone bleed, or background noise that briefly creep above the noise floor and get misclassified as tonal.

Updates in real time — no re-analysis needed. Raise the gate until breaths stop being boosted.

Situation	Suggested gate
Clean recording, quiet room	-40 dB (default)
Some breath noise	-35 to -30 dB
Loud bleed from headphones	-28 to -22 dB

Vibrato / gap tolerance (ms)

Default: 100 ms — Range: 10 to 200 ms

Maximum gap between two tonal detections that gets bridged into a single continuous segment.

A natural vibrato oscillates at 5–7 Hz, creating amplitude dips of ~70–100 ms. Without bridging, these dips create micro-segments that produce a sawtooth pattern on the envelope. Gap bridging fuses them into one smooth blob — matching Melodyne's behaviour.

Updates in real time — adjusting this slider immediately re-segments the analysis and updates the preview. No re-analysis needed.

Voice type	Suggested setting
Rap / fast lyrics	10 – 40 ms — avoid fusing separate words
Pop / rock	**80 – 120 ms (default)**
Classical / lyrical	120 – 180 ms — wide vibratos, long held notes

> ***Trade-off:*** *too high and two separate words get fused into one segment. If the singer whispers one word and belts the next 80 ms later, they'll be treated as one blob with a single averaged gain. If you hear this happening, lower the gap tolerance.*

Correction strength (%)

Default: 80% — Range: 0 to 100%

How aggressively blocks are pushed towards the target level.

- **100%** — reaches exactly the target
- **70–80%** — recommended, preserves some natural dynamics
- **50%** — gentle, acts more like a subtle tilt
- **0%** — no correction (useful to inspect segmentation only)

Transition fade (ms)

Default: 5 ms — Range: 1 to 20 ms

Duration of the gain ramp at each tonal/non-tonal boundary. Prevents clicks at abrupt gain changes. Automatically clamped to 90% of block duration.

Mode Volume Rider (per-block gain)

Default: Off (recommended)

When **off** (default): a single constant gain per segment, based on its average RMS. This is the true Melodyne equivalent — natural syllable dynamics (consonant attack, vowel decay) are preserved.

When **on**: gain recalculated every 10 ms per block's individual RMS. This is a fast volume rider / ultra-rapid compressor — flattens waveform shape and can cause pumping.

> ***Leave this off*** *for professional results. Enable only if you specifically want to flatten syllable-level dynamics.*

Smoothing (alpha) — Volume Rider mode only

Default: 0.25 — Range: 0.05 to 0.60 *(visible only when Volume Rider is on)*

Controls gain response speed between blocks. Low = smooth, slow. High = fast, more pumping risk.

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5. Step-by-step workflow

Standard procedure

1. **Select your vocal item** in REAPER

2. **Launch the script** via shortcut or Actions → Run script
3. **Open the REAPER Console** (Actions → Show REAPER console)
4. **Check the target item** is shown in green
5. **Click Analyse** — watch the progress bar and ETA
6. **Read the console:** sample rate, take offset, freq max, noise floor, block count
7. **Check the segment map:**
   - `###` = tonal regions (will be corrected)
   - `---` = noise/consonant regions (untouched)
8. **Check the auto-target:** the Target RMS slider has moved automatically to match the voice. Adjust manually if needed — use ↺ **Auto** to restore.
9. **Adjust Gate** if breaths or bleed are being boosted
10. **Adjust Gap tolerance** if vibrato sounds choppy (raise) or separate words are being fused (lower)
11. **Set correction strength** — start at 70–80%
12. **Check the estimated gain delta** in the stats panel
13. **Click Apply**
14. **Listen** — `Ctrl+Z` to undo if needed

> *The Take Volume Envelope is automatically armed on write. If the correction seems inaudible, verify the envelope is in Read mode in REAPER.*

Iterating without re-analysing

These parameters update the preview **instantly** without re-analysis:

- Target RMS level (and ↺ Auto)
- Gate threshold
- Vibrato / gap tolerance
- Correction strength
- Transition fade
- Volume Rider mode / smoothing alpha

These require **re-analysis** (blocks are re-processed):

- ZCR threshold
- Spectral centroid max
- Consonant onset sensitivity (Flux)
- Analysis resolution (block size)

Resetting the envelope

The 🗑 **Reset env.** button clears all envelope points and restores flat unity gain. Undoable with `Ctrl+Z`. Use before re-applying with different settings.

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6. Starting values by voice type

Voice type	Target RMS	Strength	ZCR	Centroid	Gap
Lead vocal — pop / rock	auto	70 – 85%	0.12 – 0.18	2000 – 2500 Hz	80 – 100 ms
Lead vocal — classical / lyric	auto	60 – 75%	0.10 – 0.15	1800 – 2200 Hz	120 – 180 ms
Rap / spoken word	auto	50 – 70%	0.18 – 0.25	2500 – 3000 Hz	10 – 40 ms
Background vocals / choir	auto	80 – 100%	0.12 – 0.18	2000 – 2500 Hz	80 – 100 ms
Acoustic / folk	auto	65 – 80%	0.10 – 0.15	1800 – 2200 Hz	80 – 120 ms
Very sibilant voice	auto	70%	0.08 – 0.10	1800 – 2000 Hz	80 ms
Breathy / airy voice	auto	75 – 85%	0.15 – 0.20	2000 – 2500 Hz	80 ms

> ***Target RMS*** *is now auto-calculated after analysis. The values above are typical ranges for reference — the auto-target will be specific to your recording level.*

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7. Where to place it in your chain

```
Recording
      |
      ▼
  [Vocal Volume Macro]   ← tonal parts levelled, noise untouched
      |
      ▼
     [De-esser]          ← sibilants intact, clean predictable target
      |
      ▼
    [Compressor]         ← input already balanced, lighter settings
      |
      ▼
      [EQ]
      |
      ▼
   [Reverb / FX]
```

Why this order:

- Script has levelled vowels → de-esser has a consistent target
- Compressor input is already balanced → less ratio, less gain reduction, less pumping
- De-esser works on untouched sibilants → more accurate detection

8. Reading the console output

```
[VVM] sr=44100 Hz | nch=2 | block=441 smp | N_eff=111 | n_bins=50 | freq_max=2480 Hz
  [VVM] Take offset: 0.0000 s | playrate: 1.0000
  [VVM] Noise floor estimate: -52.3 dB → adaptive silence threshold: -46.3 dB
  [VVM] Analysis complete — 1800 blocks.
  [VVM] Auto-target: tonal RMS = -21.4 dB → target set to -21.4 dB
  [VVM] Take offset: 0.0000 s | playrate: 1.0000
  [VVM] 312 envelope points written (decimation active).
```

Field	Meaning
`sr`	Sample rate of the audio source
`nch`	Channel count (1 = mono, 2 = stereo)
`freq_max`	Highest frequency the DFT can see (physical limit of SC slider)
`Take offset`	Seconds from file start to item start — non-zero means the item is a trim
`playrate`	Take playback rate (1.0 = normal)
`Noise floor estimate`	10th percentile RMS across all blocks
`adaptive silence threshold`	Noise floor + 6 dB — blocks below this are ignored
`Auto-target`	Tonal RMS average detected → slider moved to this value
`envelope points written`	Total after decimation (typically a few hundred for 3 min)

9. Troubleshooting

Script doesn't launch

Cause: ReaImGui not installed or outdated. **Fix:** See Installation.

"Source audio vide ou offline" error

The item's source file is missing from disk (REAPER shows "OFFLINE"), or the take is MIDI or empty. **Fix:** Locate the missing file, or select a valid audio item.

Analysis doesn't start

- Make sure an item is selected (highlighted in REAPER)
- Make sure the item is not shorter than ~100 ms

- Check the console for the exact error message

Take Envelope is not written

- Try manually: right-click item → **Take** → **Volume Envelope**
- Check the track is not write-protected
- Check the console for the error message

Correction is inaudible

Two likely causes:

1. **Audio already near the target.** If auto-target set to -21.4 dB and your source is at -21.8 dB, the correction is only +0.4 dB. Try a more extreme target to confirm the script works.
2. **Envelope not in Read mode.** The script arms it automatically, but verify the `R` status in the REAPER envelope lane.

Sibilants or breaths are being corrected

- Lower ZCR threshold (e.g. 0.15 → 0.10)
- Lower spectral centroid (e.g. 2500 → 2000 Hz)
- Lower flux onset (e.g. 0.30 → 0.20)
- **Raise Gate threshold** — most effective against breaths and bleed

Vibrato sounds choppy / sawtooth envelope

- **Raise gap tolerance** (e.g. 80 → 120 ms)
- The envelope should smooth out immediately as you drag the slider

Two separate words are being fused together

- **Lower gap tolerance** (e.g. 100 → 40 ms)
- Particularly important for rap, spoken word, or any fast-paced delivery

Result sounds robotic or over-compressed

- Make sure **Volume Rider mode is off** (most common cause)
- Lower correction strength (try 50–60%)
- Increase transition fade (try 10 ms)

Segment map shows only dashes

No tonal blocks were detected. Try:

- Raising ZCR threshold (e.g. 0.15 → 0.25)
- Raising spectral centroid (e.g. 2500 → 3000 Hz)

- Check the console for the adaptive silence threshold — if it's at -30 dB, the recording may be very quiet

Analysis is very slow

- Increase block size (20–30 ms) — fewer blocks, faster analysis
- REAPER stays responsive throughout (background defer loop)
- The ETA display gives a real-time estimate

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10. Quick reference card

Parameter	Default	Notes
ZCR threshold	0.15	Lower = stricter tonal detection
Spectral centroid	2500 Hz	Capped at freq_max after analysis
Flux onset	0.30	Lower = more plosives preserved
Block resolution	10 ms	Changing this resets the analysis
Target RMS	**auto**	Set automatically after analysis
Gate threshold	-40 dB	Raise to stop boosting breaths/bleed
Gap tolerance	100 ms	Raise for vibrato, lower for rap
Strength	80%	Avoid 100% to keep natural dynamics
Transition fade	5 ms	Raise to 8–12 ms if clicks appear
Volume Rider mode	Off	Leave off for Melodyne-style result
Smoothing alpha	0.25	Only relevant when Volume Rider is on

Buttons:

- 🔍 **Analyse** — background analysis, locks item, shows ETA, auto-sets target
- ✅ **Apply** — writes Take Volume Envelope (armed automatically, single Ctrl+Z undo)
- 🗑 **Reset env.** — clears envelope, restores 0 dB, undoable
- ↺ **Auto** — restores the auto-calculated target RMS (appears after analysis)
- ✖ **Cancel** — interrupts analysis in progress
- **Change** — unlocks item to select a different one

Parameters that update without re-analysing: Target · Gate · Gap tolerance · Strength · Fade · Mode · Alpha
Parameters that require re-analysis: ZCR · Centroid · Flux · Block resolution

> ***General advice:*** *run the analysis, let the auto-target set itself, check the segment map, adjust the gap tolerance if the map looks choppy, raise the gate if breaths appear. Apply at 70–80% strength, listen, fine-tune.* `Ctrl+Z` *is always available.*